

Ambulo - Hiking App

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Statement Of Work

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Introduction

Nature hikes are an integral part of the leisure experience for people around the world. However, planning a successful trip is a complex task. It requires information from multiple sources like maps, weather apps, and recommendation platforms. The need to consolidate this scattered information and adapt it to individual preferences makes the process time-consuming and complicated.

Additionally, hikers frequently encounter unexpected challenges during their trips, such as confusing trail splits, changing terrain conditions, or sudden weather shifts. Limited access to real-time updates can diminish the experience and, at times, pose safety risks.

The Ambulo project aims to address these issues by providing an innovative, all-in-one solution for planning and navigating hiking trips. Leveraging advanced technologies and artificial intelligence, Ambulo integrates personalized route planning, real-time navigation, and a vibrant community platform for sharing information and experiences.

The app will utilize algorithms to match routes to the user's profile, current weather conditions, and desired difficulty level. It will also offer real-time updates on terrain and trail conditions, points of interest, and tailored equipment recommendations, ensuring a seamless and enjoyable experience for hikers.

Ambulo's data will be sourced from an internal database, external APIs, and user contributions, creating a dynamic and up-to-date platform. The project is structured into three main components:

1. **Algorithm Development** – An algorithm will personalize route suggestions based on user preferences, location, and community recommendations, while also offering equipment guidance.
2. **SaaS Platform:**
A cloud-hosted interface will provide maps, real-time alerts, and community-shared reviews, ensuring seamless accessibility across devices.
3. **Backend Infrastructure:**
Centralized, scalable systems will manage data processing and feature integration, enabling efficient and secure service delivery.

Ambulo's mission is to make hiking more accessible, enjoyable, and safe for everyone, allowing users to explore and appreciate nature in a way that best suits their needs.

Goals

To create an accessible, clear, and efficient experience in trip planning and during the trip for all types of travelers, offering a variety of options based on the user and their desires.

Objectives

1. **Personalized Route Selection:** Routes will be recommended based on the user's preferences and hiking history.
2. **Comprehensive Route Descriptions:** Each route will include detailed descriptions, photos, and a list of recommended equipment.
3. **User Progression:** Users will advance through levels, unlocking new permissions, such as editing and creating routes, based on their contributions to the platform.
4. **Weather Forecast Integration:** A weekly weather forecast will be displayed, enabling users to plan ahead effectively.
5. **Seamless Login via OAuth2:** Users can quickly and securely connect to the platform using their Google accounts.
6. **Real-Time Location Features:** During hikes, the application will display the user's current location and provide context-specific alerts, such as tricky trail splits or extreme weather changes.
7. **Tailored Recommendations:** Suggestions will be dynamically generated based on the user's profile and past activities.
8. **Interactive Maps:** The application will feature user-friendly maps showing the current location and route details.
9. **Flexible Route Adjustments:** Users can easily modify routes based on changing preferences, such as proximity to their current location or filtering by region.

Metrics

Personalized Route Selection

- **Metric: Route Match Accuracy**
 - Definition: The percentage of recommended routes that align with the user's preferences (e.g., difficulty, distance, terrain type, past activity, etc).
 - Goal: $\geq 80\%$ of routes recommended should align with the user's past behavior and preferences.
 - **Metric: User Satisfaction with Recommendations**
 - Definition: Rating given by users to the quality of personalized route suggestions on a 5-point scale.
 - Goal: Average rating $\geq 4.2/5$.
-

Comprehensive Route Descriptions

- **Metric: Description Completeness - Officials**
 - Definition: The percentage of routes that include all required elements (description, photos, recommended equipment).
 - Goal: 100% of routes in the Officials should include all the required elements.
-

Weather Forecast Integration

- **Metric: Weather Data Accuracy**
 - Definition: The percentage of weather forecasts provided that are consistent with real-time weather conditions.
 - Goal: $\geq 90\%$ accuracy in weather data.
 - **Metric: User Satisfaction with Weather Forecast**
 - Definition: Rating given by users for the accuracy and usefulness of weather forecasts.
 - Goal: Average rating $\geq 4/5$.
-

Real-Time Location Features

- **Metric: Location Accuracy**
 - Definition: The percentage of locations tracked that are within a defined error margin (e.g., 10 meters).
 - Goal: $\geq 95\%$ accuracy of location tracking.
- **Metric: Real-Time Alerts Engagement**
 - Definition: Number of times users engage with real-time alerts (e.g., by tapping or acknowledging them).
 - Goal: $\geq 70\%$ engagement rate with real-time alerts.

Literature Review

Matching Algorithms

In the context of the Ambulo project, matching algorithms are essential for providing personalized hiking route recommendations based on user preferences, current conditions, and community feedback. Here, we'll explore various algorithms that can be employed to enhance the user experience.

Collaborative Filtering

Collaborative filtering is a popular recommendation technique that makes automatic predictions about user interests by collecting preferences from many users.

- **User-Based Collaborative Filtering:** Recommends routes that similar users have liked. It assumes that users with similar past preferences will have similar tastes in the future.
- **Item-Based Collaborative Filtering:** Focuses on finding similarities between routes. It recommends routes similar to those a user has liked in the past.

References:

Sarwar, B., Karypis, G., Konstan, J., & Riedl, J. (2001). Item-based collaborative filtering recommendation algorithms. *Proceedings of the 10th International Conference on World Wide Web*.

Content-Based Filtering

Content-based filtering recommends routes by comparing route attributes with a user's preferences.

- **Feature Extraction:** Identify key attributes of routes such as difficulty, length, elevation, terrain type, and points of interest.
- **Preference Profiling:** Create a user profile based on preferences and past interactions.
- **Similarity Measurement:** Calculate similarity scores between routes and user profiles using methods like cosine similarity.

References:

Lops, P., De Gemmis, M., & Semeraro, G. (2011). Content-based recommender systems: State of the art and trends. *Recommender Systems Handbook*.

Hybrid Approaches

Combining collaborative and content-based methods can offset the limitations of each.

- **Weighted Hybrid:** Integrate the scores from both collaborative and content-based algorithms with assigned weights.
- **Switching Hybrid:** Switch between algorithms based on certain criteria, such as data availability.

References:

Burke, R. (2002). Hybrid recommender systems: Survey and experiments. *User Modeling and User-Adapted Interaction*.

Implementation Strategy for Ambulo

To effectively use matching algorithms in Ambulo:

- **Data Aggregation:** Collect data from user profiles, route details, real-time weather updates, and community feedback.
- **Profile Creation:** Build dynamic user profiles that adapt to preferences and past behaviors.
- **Algorithm Selection:** Choose appropriate algorithms based on data characteristics and user needs.
- **User Feedback Loop:** Incorporate user feedback to refine recommendations and enhance accuracy.

By integrating these matching algorithms, Ambulo will offer tailored route suggestions, real-time updates, and an engaging user experience, fulfilling its mission to make hiking more accessible and enjoyable.

Competitors

Off Road

Founded in **2017**, Off Road.io provides a navigation solution for off-road enthusiasts, focusing on sharing and tracking off-road trails. The app has garnered **tens of thousands of downloads** and continues to grow in popularity, especially among 4x4 drivers.

Amud Anan (עמוד ענן)

Developed in **2014**, Amud Anan is an Israeli hiking and navigation app known for its user-contributed trail database. It serves **500,000+** of users annually, particularly hikers and bikers in Israel.

Komoot

Founded in **2010**, Komoot is a platform for outdoor activity planning, boasting over **30 million users worldwide**. It generates income through premium features, reportedly earning **millions of euros annually** through subscriptions and map packs.

AllTrails

Launched in **2010**, AllTrails has become a leading hiking app with over **40 million users globally**. The app earns an estimated **\$60 million annually**, primarily through subscriptions to its Pro features.

Category	Parameter	Ambulo	Off Road	Amud Anan	Komoot	AllTrails
Services and Features	Personalized route planning	Yes	Yes	No	Yes	Yes
	Integration of topographic maps and trails	Yes	Yes	Yes	Yes	Yes
	Live alerts for complex points	Yes	No	No	No	No
	Customized equipment lists	Yes	No	No	No	No
	Integrated weather forecasts	Yes	Yes	No	Yes (Only paid)	Yes
	External map integration (Google/OpenStreet Map)	Yes	Yes	Yes	Yes	Yes
User Experience	Simplicity and navigation process (rating from 1-3, 3 is the hardest)	1 (editable)	1 (editable)	3 (Not editable)	2 (Not editable GPX file only)	1 (Not editable GPX file only)
	Multi-language support	No (at this point)	Yes	No	Yes	Yes
Community Engagement	Route ratings	Yes (Starts + filters)	Yes (only based on stars)	No	Yes (only based on stars)	Yes (Starts + filters)

Category	Parameter	Ambulo	Off Road	Amud Anan	Komoot	AllTrails
	Adding reviews and photos	Yes	Yes	Yes	Yes	Yes
	Community-contributed routes	Yes	Yes	No (Only POI)	Yes	Yes
	System based routes	Yes	No	No	No	No
	User levels and special permissions	Yes	No	No	No	No
Technological Innovation	Use of personalized algorithms	Yes	Yes	No	Yes	Yes
	Real-time data integration	Yes	Yes	No	Yes (only paid)	Yes
	Variety of map types	Yes	Yes	Yes	Yes	Yes
Business Model	Free vs. paid versions	Free (at this point)	Free (some maps are paid)	Free (All offline maps are paid)	Free (Most features are paid only)	Mostly paid
	Integrated Ads	Minimal	Minimal	No	Minimal	Highly
	Partnerships (attractions, restaurants)	Not at this point	No	No	Yes	Yes
Support and Maintenance	Regular updates and improvements	Yes	Yes	Minimal (look the same since 2014)	Yes	Yes

detailed requirements

Identify Stakeholders

- **End-Users (Hikers):**
Individuals who use the Ambulo app to plan and navigate their hiking trips.
- **Project Managers:**
Oversee project development, ensuring milestones and objectives are met.
(No GUI)
- **Developers:**
Responsible for building, maintaining, and enhancing the Ambulo application.
- **Designers:**
Create the user interface and ensure a seamless user experience.
(No GUI)
- **System Administrators:**
Manage backend infrastructure, databases, and ensure system reliability.
- **External Partners:**
Providers of third-party services such as weather data APIs and mapping services.
(No GUI)
- **Investors and Stakeholders:**
Individuals or organizations funding the project and expecting returns or specific outcomes.
(No GUI)
- **Regulatory Bodies:**
Ensure the application complies with relevant laws and regulations, such as data protection standards.
(No GUI)

Functional Requirements

1. User Authentication and Profiles

Profile Management:

Store user preferences, hiking history, and contribution records.

Allow users to update their profile information, including personal details and hiking preferences.

OAuth2 Integration:

Allow users to register and log in securely using their Google accounts via OAuth2.

2. Personalized Route Planning

Route Recommendations:

Utilize algorithms to suggest hiking routes based on user preferences, location, hiking history, and desired difficulty level.

Search and Filter:

Enable users to search for routes by

- | | |
|--------------------------------|-----------------------|
| - Distance | - Payment required |
| - Region | - Recommended season |
| - Loop | - Surface type |
| - Includes water sections | - Ending point |
| - Number of nights | - Estimated time |
| - Trail type, Difficulty level | - Wheelchair Friendly |
| - Starting point | - Dog Friendly |

Save and Modify Routes:

Allow users to save favorite routes to their profiles.

Enable users to modify saved routes based on changing preferences or conditions.

3. Comprehensive Route Descriptions

Detailed Information:

Provide comprehensive details for each route, including trail length, difficulty level, elevation gain, and estimated hiking time.

Multimedia Content:

Include photos and maps to give users a visual understanding of the route.

Points of Interest:

Highlight significant landmarks, scenic viewpoints, and rest areas along the route.

Equipment Recommendations:

Suggest necessary equipment based on route difficulty and weather conditions.

4. Interactive Maps

Real-Time Location Display:

Show the user's current location on the map during hikes.

Route Visualization:

Display the entire hiking route with markers for key points of interest and trail splits.

Map Layers:

Provide options to toggle different map layers such as terrain, satellite view, and trail conditions.

Zoom and Pan:

Allow users to zoom in and out and navigate around the map seamlessly.

5. Real-Time Alerts

Trail Condition Updates:

Provide real-time information on trail conditions, including closures, hazards, and maintenance activities.

Weather Alerts:

Notify users of sudden weather changes or extreme conditions during hikes.

6. Weather Forecast Integration

Weekly Forecasts:

Display detailed weekly weather forecasts to help users plan their hikes effectively.

Real-Time Weather Updates:

Integrate with external weather APIs to provide up-to-date weather information.

7. Community Features

User Reviews and Ratings:

Allow users to leave reviews and rate hiking routes based on their experiences in the community section.

Photo Sharing:

Enable users to upload and share photos from their hikes.

Trail Suggestions:

Let users suggest new trails and provide detailed information for inclusion in the platform.

8. Equipment Recommendations

Tailored Suggestions:

Recommend equipment based on the selected route's difficulty

Customizable Lists:

Allow users to create and customize equipment lists for specific hikes. There will be an option to create a group with other users where the entire trip, including planning, equipment, and more, will be shared among them.

9. Flexible Route Adjustments

Dynamic Modifications:

Enable users to adjust their hiking routes in real-time based on changing preferences or unexpected conditions.

Proximity-Based Filtering:

Allow users to filter and adjust routes based on their current location.

Regional Filtering:

Provide options to filter routes by specific districts or areas of interest.

10. Real-Time Location Features

GPS Integration:

Utilize device GPS to track and display the user's real-time location on the map.

Context-Specific Alerts:

Provide alerts related to the user's current location, such as upcoming trail splits or areas prone to sudden weather changes.

Navigation Assistance:

Offer turn-by-turn navigation and route guidance to help users stay on track.

11. Data Synchronization and Offline Access

Sync Across Devices:

Ensure user data is synchronized across multiple devices for a consistent experience.

Offline Mode:

Allow users to access maps and route information offline, with data synchronization once back online.

Data Caching:

Implement caching mechanisms to store essential data locally for improved performance and reliability.

Non-Functional Requirements

1. Performance

Throughput:

The system should handle a minimum of 1000 concurrent users without significant performance degradation.

Resource Utilization:

Optimize server and database resources to handle high traffic efficiently.

2. Scalability

Horizontal Scaling:

Design the system to scale horizontally by adding more instances of each microservice to accommodate increasing user loads.

3. Reliability and Availability

Uptime:

Achieve an availability target of 99.9%, allowing for scheduled maintenance windows.

Redundancy:

Implement redundant servers and failover mechanisms to minimize downtime in case of component failures.

Disaster Recovery:

Establish disaster recovery plans, including regular data backups and failover strategies.

4. Security

Authentication and Authorization:

Implement secure OAuth2 authentication and role-based access control (RBAC) to manage user permissions.

Vulnerability Management:

Conduct regular vulnerability scanning and penetration testing using tools like

5. Usability

User Interface:

Design an intuitive and user-friendly interface with clear navigation and accessible design.

Accessibility:

Adhere to WCAG standards to ensure the application is accessible to users with disabilities.

Responsive Design:

Ensure the application is fully responsive and functions seamlessly across various devices and screen sizes.

6. Maintainability

Code Quality:

Write clean, modular, and well-documented code to facilitate easy maintenance and updates.

Documentation:

Maintain comprehensive documentation for code, APIs, and system architecture using tools like JSDoc and Swagger.

Version Control:

Use Git with GitHub for effective source code management and collaboration.

7. Compatibility

Browser Support:

Ensure compatibility with major browsers, including Chrome, Firefox, Safari, and Edge.

Platform Support:

Support both web and mobile platforms with consistent functionality and performance.

Device Compatibility:

Ensure the application works seamlessly across various devices, including desktops, tablets, and smartphones.

8. Reliability

Error Handling:

Implement robust error handling to gracefully manage and recover from system failures.

Automated Alerts:

Set up automated alerts for critical system events and performance issues to enable timely responses.

9. Performance Optimization

Database Optimization:

Optimize database queries and use indexing to enhance data retrieval performance.

Load Balancing:

Implement load balancing to distribute traffic evenly across servers, ensuring efficient resource utilization.

10. Data Management

Data Integrity:

Maintain data integrity through validation and consistent data handling practices.

Data Backup:

Perform regular data backups and implement retention policies to prevent data loss.

11. Testing and Quality Assurance

Automated Testing:

Implement automated testing to ensure code reliability and functionality.

System Architecture

SaaS Architecture:

Client Interface: Web and mobile applications offering intuitive user experiences for hikers.

Cloud Backend: Centralized servers handle business logic, data processing, and API requests for seamless operation.

Databases: Firebase handles real-time data storage, structured data, and user-generated content.

APIs: RESTful APIs enable smooth communication between the interface and backend systems.

Technological Requirements

Define key technical requirements:

Browsers: Chrome, Firefox, Edge.

Devices: Desktop, tablet, mobile (Android).

Operating Systems: Windows, Linux, Android.

Frameworks: Flutter for both frontend and backend.

Infrastructure Requirements:

Hosting Platforms: AWS or Azure for backend services.

Servers: Utilize scalable cloud servers with auto-scaling capabilities.

Databases: PostgreSQL for relational data, FireBase for NoSQL data.

Caching: shared preferences from pub.dev for in-memory caching to enhance performance.

Database Technology:

NoSQL DBMS: Firebase for flexible and real-time data storage.

Scalability: Implement sharding and replication for database scalability.

Cloud Services:

Deployment: AWS Elastic Beanstalk or Azure App Services.

Storage: AWS S3 for user-uploaded content.

CI/CD: Jenkins for automated testing and deployment.

Development Tools:

Version Control: Git with GitHub.

IDE: Visual Studio Code.

Use Cases

Use Case 1: User Registration and Profile Management

Actor: New User

Goal: Register for an account and set up a personalized profile.

Steps:

Registration:

User navigates to the Ambulo app's registration page.
User selects "Sign up with Google" to initiate OAuth2 authentication.
Users grant necessary permissions for Ambulo to access their Google account information.
System creates a new user account and redirects the user to the profile setup page.

Profile Setup:

User input additional information such as hiking preferences, favorite trail types, and personal details.
User saves the profile information.

Possible Results:

Success: User successfully registers and sets up their profile, gaining access to personalized features.

Failure: If OAuth2 authentication fails, the system notifies the user and prompts retrying the registration process.

Use Case 2: Plan a Personalized Hiking Trip

Actor: Registered User

Goal: Create and save a personalized hiking trip based on preferences and current conditions.

Steps:

Input Preferences:

User access the "Plan a Trip" feature in the Ambulo app.

User selects preferences such as desired location, difficulty level, trail length, and preferred hiking season.

Receive Recommendations:

System processes the input using recommendation algorithms.

System displays a list of suggested hiking routes that match the user's criteria and current weather conditions and other information.

Review and Select Route:

Users review detailed descriptions, photos, and equipment recommendations for each suggested route.

User selects a preferred route and there will be two option - start, save

Save and Modify:

System saves the selected route to the user's favorite.

Users can later modify the saved route based on changing preferences or new conditions.

Possible Results:

Success: User successfully plans and saves a personalized hiking trip.

Failure: No routes match the user's criteria, prompting the user to adjust their preferences and suggesting alternative routes

Use Case 3: Navigate a Hike with Real-Time Alerts

Actor: Active Hiker

Goal: Navigate a hiking trail with real-time updates and alerts for safety and convenience.

Steps:

Start Navigation:

User selects a saved hiking route and clicks "Start Hike."
System activates GPS tracking and displays the current location on the interactive map.

Receive Real-Time Alerts:

As the hike progresses, the system sends real-time alerts about trail conditions, weather changes, and upcoming tricky trail splits.
Users receive push notifications and in-app alerts based on their location and predefined preferences.

Emergency Assistance:

In case of an emergency, user can access quick links to emergency contacts and safety instructions directly from the alerts.

Possible Results:

Success: User navigates the trail safely with timely alerts and assistance.

Failure: If GPS signal is lost, the system notifies the user and attempts to reconnect.

Use Case 4: Share Hiking Experience in Community

Actor: Registered User

Goal: Share reviews, photos, and suggestions to contribute to the Ambulo community.

Steps:

Create a Review:

After completing a hike, the user accesses the "Community" section.
The user selects the completed route and writes a review detailing their experience, including ratings for difficulty, scenery, and trail conditions.

Upload Photos:

The user upload photos taken during the hike to accompany the review.
Photos are tagged with location and time for better context.

Suggest a New Trail:

The user selects the option to suggest a new trail if they explored an unmapped area. The user provides detailed information about the new trail, including trailhead location, length, difficulty, and points of interest.

Engage with Community:

Other users can view, like, and comment on the reviews and photos.
User receives.

Possible Results:

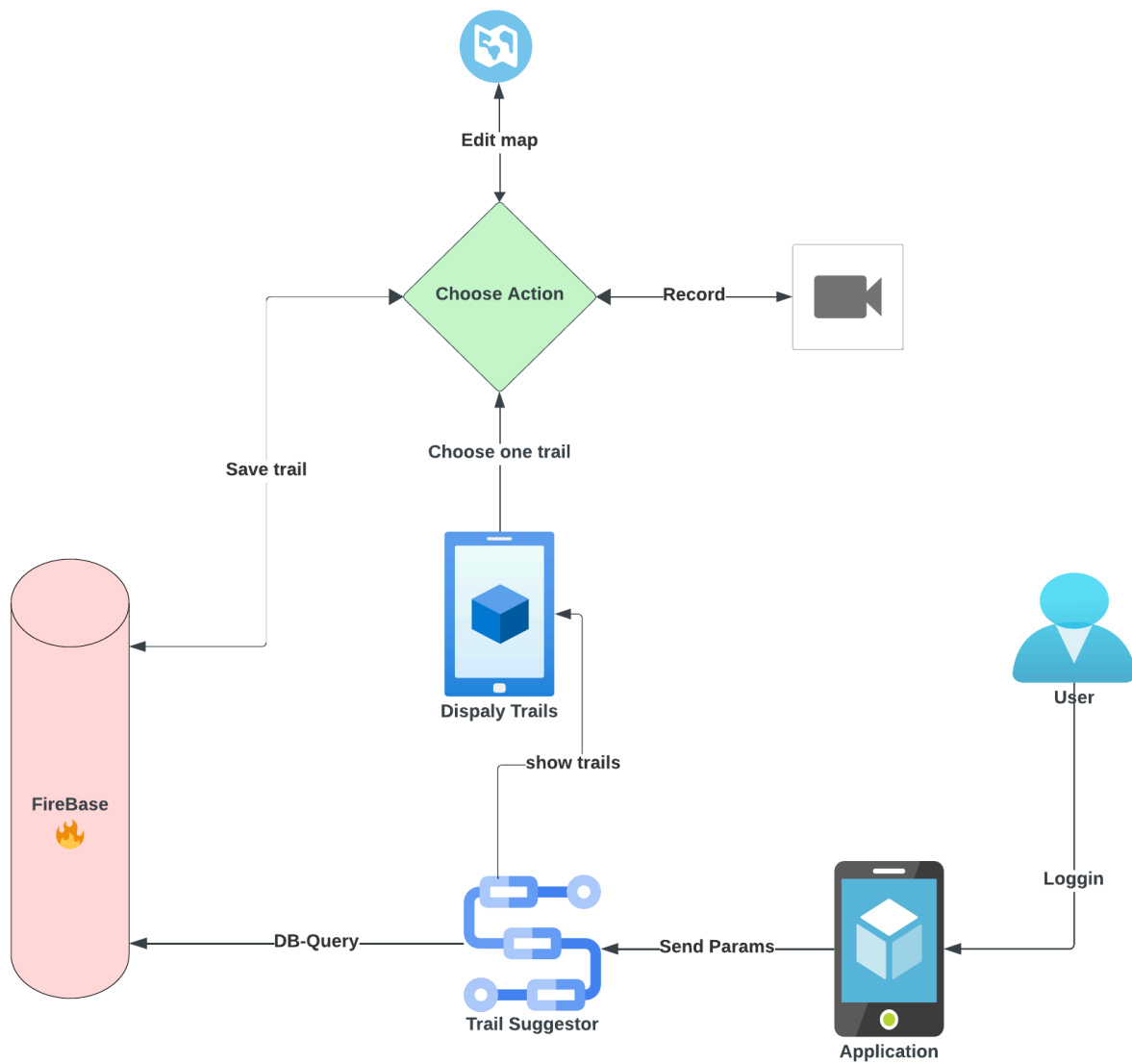
Success: Users successfully share their hiking experience, contributing valuable information to the community.

Failure: If the suggested trail lacks necessary details, the system prompts the user to provide additional information.

Detail Design

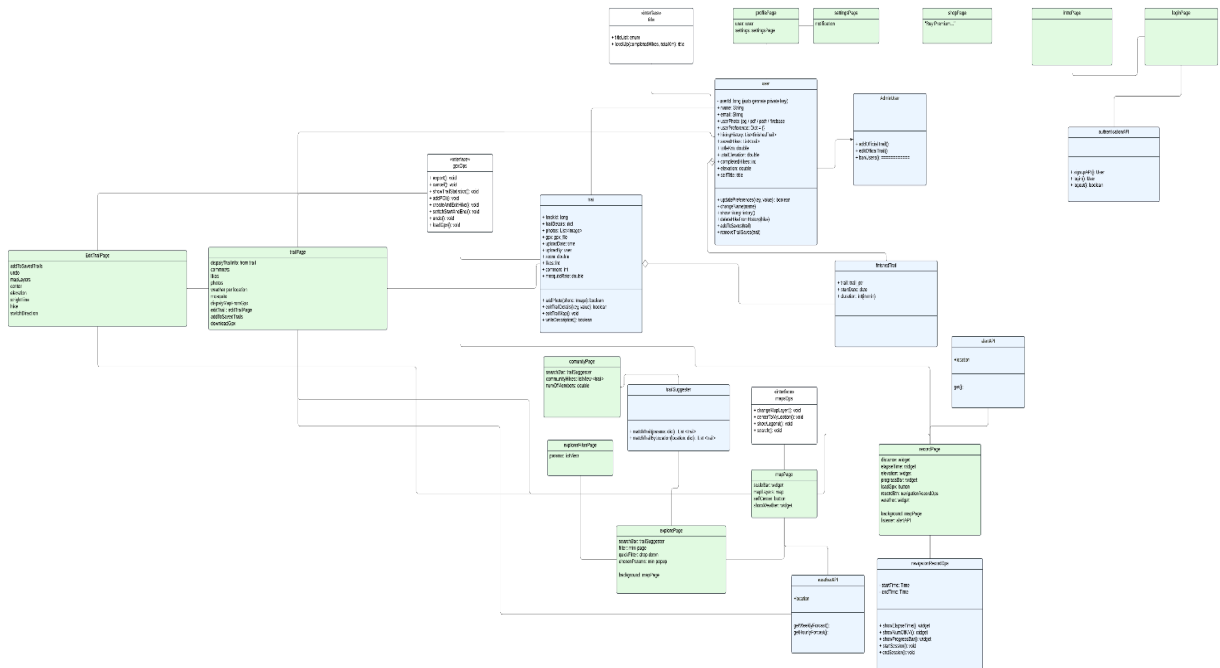
High-level Architecture

[Link to the SVG](#)



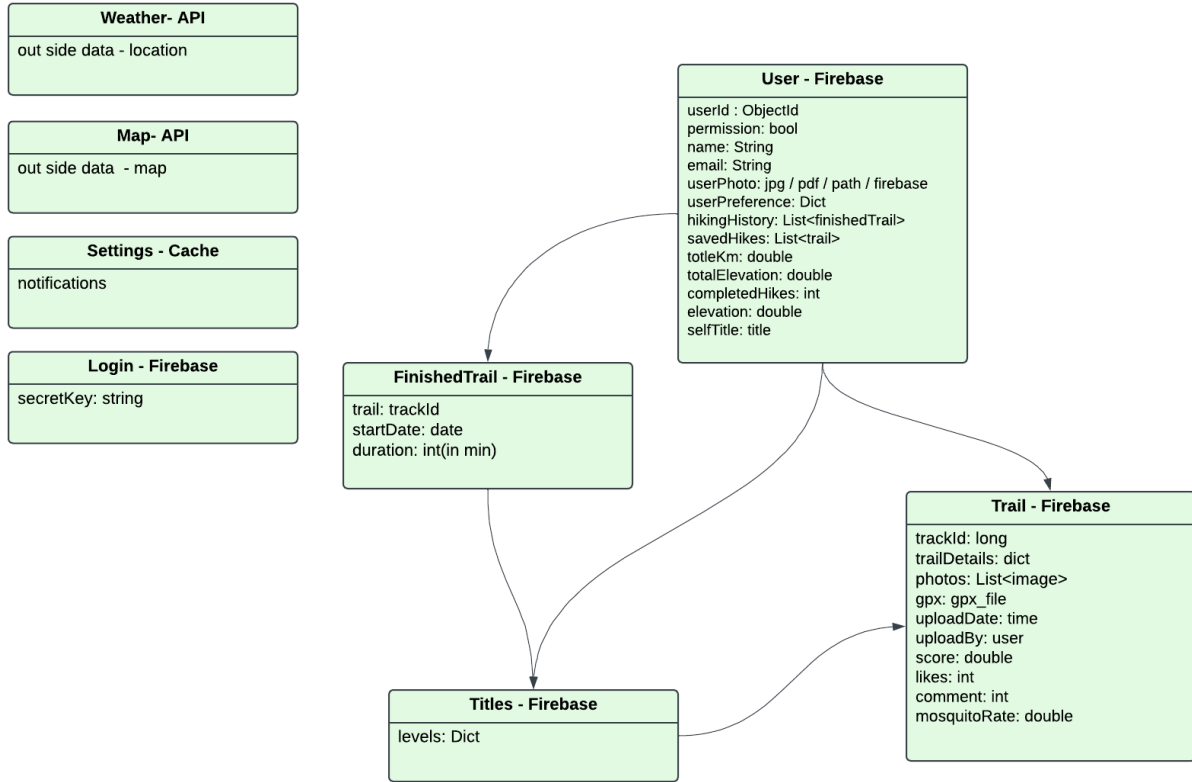
UML - Class

[Link to the SVG](#)



DataBase Schema

[Link to the SVG](#)



Development stages

Task name	Start	End
Presentation	23/12/2024	27/12/2024
Presentation - class	16/1/2025	16/1/2025
Code	25/1/2025	25/3/2025
Test the app	26/3/2025	10/4/2025
Deployment	11/4/2025	25/4/2025